Forecast-Driven Model Predictive Control with Reinforcement Learning for Low-Carbon Community Battery Scheduling

1st Elena Margaret Ellis
School of Computer Science and Engineering
Vellore Institute of Technology
Chennai, India
elenamargaret.ellis2023@vitstudent.ac.in

2nd Anora V
School of Computer Science and Engineering
Vellore Institute of Technology
Chennai, India
anora.v2023@vitstudent.ac.in

3rd Cherish Vasant
School of Computer Science and Engineering
Vellore Institute of Technology
Chennai, India
cherish.vasant2023@vitstudent.ac.in

4th Dr. Natarajan B
School of Computer Science and Engineering
Vellore Institute of Technology
Chennai, India
natarajan.b@vit.ac.in

***Abstract*—Residential batteries can reduce electricity costs and emissions by shifting consumption in time, but a controller in a real building cannot observe future demand, solar generation or carbon intensity. This work proposes an RL-tuned model predictive controller for the batteries of a whole district. Gradient-boosted forecasters predict the load and solar output of every building, and a linear program uses these forecasts to plan all batteries over the next 30 hours. A reinforcement learning (RL) agent, trained with Truncated Quantile Critics (TQC) and a reward that adds up to the official performance indicators, sets three weights of this program every hour from district-level features. On the CityLearn Challenge 2022 benchmark, with 17 buildings and three seeds, the controller lowers the electricity cost by 34.3 percent and the emissions by 22.2 percent in the training district. Its mean Phase III public score is 0.774 and its private score 0.844, which places it 8th among the 42 public entries, and its carbon score of 0.764 is the lowest of all 42 entries. Our ablations show that the optimiser, the forecasts and the RL tuning all contribute.**

***Index Terms*—Battery energy storage, demand response, deep reinforcement learning, model predictive control, truncated quantile critics, load forecasting, reward design, CityLearn, carbon-aware control, grid-interactive buildings.**

I. Introduction

Buildings use a large share of all electricity, and rooftop photovoltaic (PV) panels and home batteries now turn many of them into flexible grid resources. Batteries that charge in low-price or low-carbon hours and discharge in high-price or carbon-intensive hours reduce both cost and emissions, and across a district they can smooth the aggregate load seen by the grid [1], [2]. The resulting control problem is challenging, because each action changes the state of charge and its effect extends many hours into the future, which makes credit assignment difficult.

Rule-based controllers are simple, but they cannot adapt to the weather, the tariff or the state of the grid. Model predictive control (MPC) plans ahead with forecasts, but its performance depends on the accuracy of its forecasts and on fixed weights that trade cost against emissions and grid impact. Model-free deep reinforcement learning (RL) [3] learns a policy from interaction alone [4], [5], but on its own it must discover the daily patterns of price and demand, and the long-term effect of each charge decision, from a scalar reward. The CityLearn environment [6], [7] and the NeurIPS 2022 CityLearn Challenge [8] turned this problem into a shared benchmark, whose score combines the electricity cost, the carbon emissions and two grid key performance indicators (KPIs).

This work combines the two ideas in an *RL-tuned MPC* (Fig. 1). Gradient-boosted models forecast the load and PV of each building, a linear program turns these forecasts into a battery schedule that respects the battery limits, and a TQC agent sets three weights of the program every hour. Since its inputs are district-level, one policy runs on districts of any size. Our second contribution is a *score-aligned counterfactual reward*, whose terms add up to the official KPIs, use the no-battery run as their scale and measure the gain over idle batteries. Third, we evaluate the controller on all 17 challenge buildings with the official formulas over three seeds, compare it with all 42 public leaderboard entries and the private medians, ablate each component and show which forecast errors limit the score.

II. Literature Review

A. RL for Building Energy and CityLearn

Vázquez-Canteli and Nagy [1] found that most RL studies for demand response used simple function approximators, single buildings and custom simulators, which made their results difficult to compare. Later surveys [2], [4] named sample efficiency, safety, reward design and transfer across buildings as the main open problems, and Nagy *et al.* [5] stressed the need for shared benchmarks and KPIs. Dulac-Arnold *et al.* [9] listed the challenges of real-world RL, of which multi-objective rewards, partial observability and limited data are central to battery control. CityLearn [6], [7] provides districts of buildings with storage and PV. MARLISA [10] coordinated several agents through sequential action selection, and Nweye *et al.* [11] studied the real-world challenges of multi-agent RL in such communities.

B. Forecast-Driven Optimisation in the CityLearn Challenge

MPC solves an optimisation problem over a moving horizon of forecasts and applies only the first decision [12]. According to the organisers' overview [13], all top entries of the 2022 challenge combined forecasting with optimisation or learning. The winner, Together, forecast load and PV with gradient-boosted trees and linear models and mixed a rolling-horizon stochastic optimiser with multi-agent PPO. The runner-up used dynamic programming in the first phase and, later, a small policy network trained by CMA-ES. CUFE solved a 24-hour linear program per building, while DivMARL averaged a shared SAC policy with a rule driven by XGBoost forecasts.

C. Deep RL Algorithms and Reward Design

The MPC weights are continuous, so actor-critic methods are a natural choice. DDPG [14] and TD3 [15] learn deterministic policies with off-policy critics, SAC [16] adds an entropy term with automatic temperature tuning, and PPO [17] is a common on-policy alternative. We use TQC [18], which extends SAC with distributional critics whose largest quantiles are dropped to reduce over-estimation, as implemented in Stable-Baselines3 [19]. Battery control has several conflicting objectives, so a single reward needs weights and scales for them. Ng *et al.* [20] showed which reward transformations leave the optimal policy unchanged, which our counterfactual baseline relies on, and Cao *et al.* [21] showed that a degradation model in the reward changes the learned arbitrage strategy. Because deep RL results vary strongly with the random seed [22], [23], we report the mean and standard deviation of three seeds.

III. Problem Formulation

A. Simulation Environment and Observations

We use CityLearn 2.1.2 with the data of the 2022 challenge, which contains 17 buildings in three districts: five public training buildings (1–5), five validation buildings (6–10) and seven test buildings (11–17). Each building has a load it cannot shift, rooftop PV and one battery of 6.4 kWh and 5 kW. The simulation runs hourly for one year (8,759 steps), and every hour the controller sets the charge or discharge rate of each battery. Electricity is billed under a time-of-use tariff, and the carbon intensity of the grid varies over the day. The controller observes the calendar, the weather with the environment's 6, 12 and 24-hour forecasts, the carbon intensity, the price with its tariff values, and the load, PV output, state of charge (SoC) and net consumption of each building.

B. Key Performance Indicators

Let E_t be the district's net consumption at hour t (positive for imports), π_t the price and I_t the carbon intensity. We mark the same quantity in the run without batteries with a superscript "nb". The official KPIs are

$$C = \frac{\sum_t \max(0, E_t \pi_t)}{\sum_t \max(0, E_t^{\mathrm{nb}} \pi_t)}, \qquad G = \frac{\sum_t \max(0, E_t I_t)}{\sum_t \max(0, E_t^{\mathrm{nb}} I_t)}, \tag{1}$$

$$R = \frac{\sum_t |E_t - E_{t-1}|}{\sum_t |E_t^{\mathrm{nb}} - E_{t-1}^{\mathrm{nb}}|}, \qquad D = \tfrac{1}{2}\big(R + (1 - L)\big), \tag{2}$$

where C is the cost, G the carbon emissions, R the ramping and $1-L$ the mean monthly value of $1-\bar{E}/\max E$ relative to no battery. Exports are not paid, and since the clipping applies to the district total, one building's export can offset another's import. All KPIs are ratios to the no-battery run, so values below one are gains.

The score of a district d is $S_d = (C + G + D)/3$. The first phase of the challenge ranked entries by $(C + G)/2$ on the training buildings, which we call the Phase I score. The final phase ranked them by a *public* score $0.4\,S_{\mathrm{train}} + 0.6\,S_{\mathrm{val}}$, and it chose the winners by a *private* score $0.2\,S_{\mathrm{train}} + 0.3\,S_{\mathrm{val}} + 0.5\,S_{\mathrm{test}}$ [13]. The public score is our main measure.

IV. Proposed Methodology

Every hour, the controller in Fig. 1 forecasts the next $H = 30$ hours, the RL agent sets three weights from a district-level state, and a linear program, which we call the solver, uses these weights to plan all batteries over the horizon.

A. Forecasting Module

Each hour we forecast every building's load and PV output, the carbon intensity and the price for the next H hours. One LightGBM model [24] per quantity covers all horizons and buildings. Its inputs are recent and lagged values, the mean of the same hour over the past seven days, the calendar, the current weather and the environment's weather forecast for the target hour. We divide each value input by the building's mean absolute value over the past week, so that a model trained on some buildings also applies to others. The model for a district is trained on the buildings of the other two districts, and Optuna [25] chooses its settings by cross-validation between them. Carbon intensity is forecast as the mean of the same hour yesterday and over the last week plus the current error, decaying as 0.8^h, and the price as the tariff of the same hour one week earlier.

B. Receding-Horizon Linear Program

Our MPC receives the forecast net load of the district $\hat{B}_h$ (load minus PV), the prices $\hat{\pi}_h$ and the carbon intensities $\hat{I}_h$. It then solves

$$\begin{aligned}
\min \; & \sum_{h=1}^{H} \Big(\tfrac{\hat{\pi}_h}{\bar{c}} + w_g \tfrac{\hat{I}_h}{\bar{g}} \Big) m_h + \tfrac{w_r}{2\bar{r}} \sum_{h=1}^{H} \rho_h - v\,\eta \sum_b s_{b,H} \\
\text{s.t. } & m_h \ge \hat{B}_h + \textstyle\sum_b (x^+_{b,h} - x^-_{b,h}),\; m_h \ge 0, \\
& \rho_h \ge |E_h - E_{h-1}|,\; E_h = \hat{B}_h + \textstyle\sum_b (x^+_{b,h} - x^-_{b,h}), \\
& s_{b,h} = s_{b,h-1} + \eta x^+_{b,h} - x^-_{b,h}/\eta,\; 0 \le s_{b,h} \le 6.4, \\
& 0 \le x^{\pm}_{b,h} \le \min(5,\; 21 - 3.125\, s_{b,h-1}),
\end{aligned} \tag{3}$$

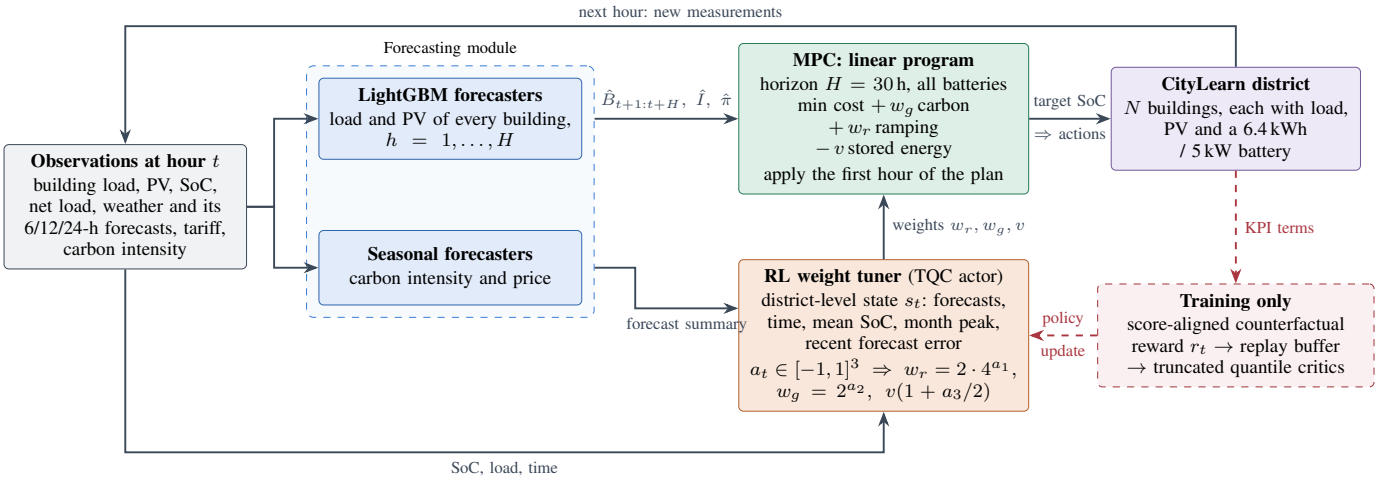


Fig. 1. Architecture of the proposed RL-tuned MPC. Every hour, the forecasters predict the next H hours. The RL agent maps a district-level state to three weights of a linear program, and the first hour of the program's plan is applied to the batteries. Dashed parts are used only during training.

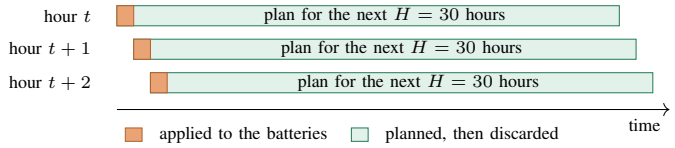


Fig. 2. Receding horizon. Every hour the linear program plans 30 hours ahead with fresh forecasts and weights, and only the first hour is carried out.

where m_h is the district import (exports are not paid), ρ_h the ramp, $s_{b,h}$ the energy stored in battery b, and $x^{\pm}_{b,h}$ the charge and discharge powers, limited to 5 kW, with an efficiency of $\eta = \sqrt{0.9}$ each way. The last constraint models CityLearn's power taper above 80% charge. The running no-battery means $\bar{c}$, $\bar{g}$ and $\bar{r}$ put cost, carbon and ramping on the scale of the KPIs, and v, the median unit cost over the horizon, values the energy left at its end. Only the first hour of the plan is applied, and the problem is solved again one hour later with new forecasts (Fig. 2). With the HiGHS solver, each problem takes about 20 ms.

C. Reinforcement Learning Weight Tuner

Fixed weights in (3) cannot adapt to the season, the forecast quality or the battery state, so an RL agent outputs three numbers $a_t \in [-1, 1]^3$ every hour. They set the ramp weight $w_r = 2 \cdot 4^{a_1}$, the carbon weight $w_g = 2^{a_2}$ and the end-of-horizon value $v(1 + a_3/2)$. Each weight is centred on a good fixed setting. A zero action thus reproduces a fixed-weight MPC. The agent's state holds the forecasts of net load, carbon intensity and price over the horizon, the time of day and season, the mean state of charge, the current load and month peak, and the latest forecast error. None of these depends on the number of buildings. A policy trained on five buildings can therefore control the seven test buildings as well.

The agent learns from a *score-aligned counterfactual reward*, built so that its terms add up exactly to the numerators of the official KPIs, on their scale. At hour t we use the district's clipped cost and emissions and its ramp,

$$\hat{c}_t = \frac{\max(0, E_t \pi_t)}{\bar{c}^{\,\mathrm{nb}}}, \qquad \hat{g}_t = \frac{\max(0, E_t I_t)}{\bar{g}^{\,\mathrm{nb}}}, \qquad \hat{r}_t = \frac{|E_t - E_{t-1}|}{\bar{r}^{\,\mathrm{nb}}}, \tag{4}$$

where a bar marks the mean of the term per step in the no-battery run. Then $\sum_t \hat{c}_t = T \cdot C$, so the return over a year is a linear function of the KPIs. The peak term $\hat{p}_t$ is the rise of the running daily maximum of E_t. The first hour of each day counts its full load, so the terms of a day add up to that day's peak. The wear term is $\hat{b}_t = \sum_i |\mathrm{SoC}_{i,t} - \mathrm{SoC}_{i,t-1}|$. The reward is

$$r_t = -\left(\hat{c}_t + w_g \hat{g}_t + w_r \hat{r}_t + w_p \hat{p}_t + w_b \hat{b}_t\right) + \left(\hat{c}_t^{\,\mathrm{nb}} + w_g \hat{g}_t^{\,\mathrm{nb}} + w_r \hat{r}_t^{\,\mathrm{nb}} + w_p \hat{p}_t^{\,\mathrm{nb}}\right). \tag{5}$$

The second bracket holds the same terms for the case in which the batteries stayed idle in that hour. Since it depends only on outside data, and not on the actions or the battery state, it leaves the optimal policy unchanged [20]. It does, however, remove most of the variation caused by demand. The agent is thus rewarded for the *improvement* it causes, and a reward of zero means "as good as no battery". A real system can compute this baseline online from the measured net load and the battery's own power.

The weights follow the score $S = (C + G + (R + 1 - L)/2)/3$: cost has weight 1, ramping and peak $w_r = w_p = 0.5$ (the peak term stands in for the load factor), wear $w_b = 0.1$, and carbon $w_g = 2$, which puts extra weight on emissions. We train the agent with TQC [18], using an actor and two distributional critics with 25 quantiles each, of which the two largest are dropped when the target is formed. This limits the over-estimation that a noisy, long-horizon reward invites.

TABLE I
THE PROPOSED RL-TUNED MPC IN EACH DISTRICT. INDICATORS RELATIVE TO NO BATTERY (LOWER IS BETTER), MEAN OF THREE SEEDS, ± STANDARD DEVIATION FOR THE SCORES. THE POLICY IS TRAINED ON THE TRAINING BUILDINGS ONLY.

District	Controller	C	G	D	Phase I	S
Train (1–5)	No battery	1.000	1.000	1.000	1.000	1.000
	RBC	1.070	1.193	1.075	1.132	1.113
	RL-tuned MPC	**0.657**	**0.778**	**0.900**	**0.717**±.002	**0.778**±.002
Validation (6–10)	RL-tuned MPC	0.655	0.755	0.904	0.705	0.771±.003
Test (11–17)	RL-tuned MPC	0.801	0.959	0.982	0.880	0.914±.006
Phase III public score $0.4\,S_{\text{train}} + 0.6\,S_{\text{val}}$						0.774 ± 0.003
Phase III private score $0.2\,S_{\text{train}} + 0.3\,S_{\text{val}} + 0.5\,S_{\text{test}}$						0.844 ± 0.004

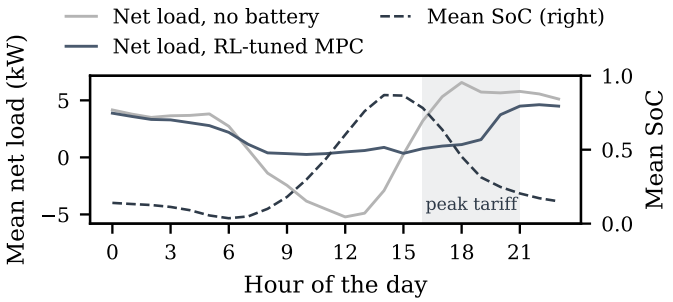


Fig. 3. Average daily profile over the year on the training buildings (seed 42): district net load with and without batteries, and the mean state of charge.

V. EXPERIMENTAL SETUP

We train the tuner for 100,000 environment steps, about 11.4 simulated years, on the five training buildings, with a learning rate of $3 \cdot 10^{-4}$, a replay buffer of 200,000 transitions and a batch size of 256. We use $\tau = 0.005$, $\gamma = 0.99$, one gradient step per environment step, 1,000 warm-up steps and automatic entropy tuning. The actor and each critic have two hidden layers of 64 units, and we run three seeds (42, 100 and 2026) and report the mean and standard deviation.

The trained policy then runs deterministically on each district for the full year without retraining. Because CityLearn 2.1.2 contains all 17 buildings, we compute the KPIs and the public and private scores ourselves with the official formulas, not with the challenge's evaluator. As references we use no battery, where every KPI equals one, and the rule-based controller (RBC) supplied with CityLearn. In the ablations, we replace one part of the controller at a time: fixed weights instead of the tuner (ramp weight 3, carbon weight 1, 24-hour horizon), horizons of 24 and 36 hours, default LightGBM settings, seasonal load and PV forecasts, SAC instead of TQC, and a TQC agent with the same reward that sets the battery rates directly, with past-only history features and 200,000 steps. Ablations with one seed use seed 42.

VI. RESULTS

A. Performance Across Districts

The controller improves every KPI in every district (Table I). On buildings 1–5 it lowers the cost by 34.3% and the emissions by 22.2% relative to no battery and smooths the district load ($R = 0.823$, $1 - L = 0.977$, $D = 0.900$), whereas the RBC performs worse than idle batteries on every KPI. The gains transfer to the validation district, where the cost falls by 34.5% and the emissions by 24.5%. The scores of the training and validation districts are almost equal ($S = 0.778$ and 0.771), so the controller is not fitted to the buildings it was trained on. The seven test buildings differ most from the training ones and gain less ($S = 0.914$), with the cost down by 19.9% and the emissions by 4.1%. The standard deviation across seeds is at most 0.006.

B. Analysis of the Learned Policy

We first analyse the learned policy through the average daily profile in the training district (Fig. 3). With no batteries, the district exports up to 5 kW of PV at midday, which is not paid, and imports about 6 kW in the evening. The controller stores this surplus. The mean state of charge rises from under 10% at dawn to about 87% at 14:00, and the batteries then discharge through the peak tariff from 16:00 to 21:00 and into the night, which gives a considerably flatter load at the grid connection. On some days, however, the controller charges from the grid in the last off-peak hour before the peak tariff. These short import spikes add little to C and G but raise the ramping and the monthly peak, which is the main reason why D is our weakest KPI.

C. Ablation Study

Table II changes one component at a time. Removing the linear program causes by far the largest drop in performance: an RL agent with the same reward that sets the battery rates directly reaches only $S = 0.862$ on buildings 1–5 and a public score of 0.854, even with history features and twice the training, and it ramps more than idle batteries do ($D = 1.06$). With fixed weights instead of the tuner, the controller scores 0.781 (public) and 0.846 (private) against 0.774 and 0.844, and its Phase I score rises from 0.717 to 0.742, mainly through cost (0.703 against 0.657).

The benefit of the tuner depends on the forecast accuracy. With seasonal load and PV forecasts, the controller still performs well in the training district (Phase I 0.728) but loses most of its advantage on unseen buildings (public 0.807, private 0.864), because the agent overfits its weights to the training buildings. Tuning the LightGBM settings raises the PV accuracy from about 90% to 96% and lowers the public score from 0.782 to 0.774. The horizon and the RL algorithm have a much smaller effect: horizons of 24, 30 and 36 hours give public scores within 0.002 of each other, and SAC in place of TQC gives 0.775 ± 0.001.

D. Comparison with the Challenge Leaderboards

On the public leaderboard of the final phase [8], which reports each entry's C, G and D as a 0.4/0.6 blend of the training and validation districts, our score of 0.774 would rank

TABLE II
ABLATIONS OF THE RL-TUNED MPC, ONE CHANGE AT A TIME (LOWER IS BETTER). PHASE I AND S ON THE TRAINING BUILDINGS, S ON THE UNSEEN VALIDATION AND TEST BUILDINGS, AND THE PHASE III SCORES. THREE SEEDS WHERE MARKED *, OTHERWISE SEED 42.

Variant	Phase I	S_{train}	S_{val}	S_{test}	Public	Private
Proposed (TQC, tuned LightGBM, 30 h)*	0.717	0.778	**0.771**	0.914	**0.774**	**0.844**
Fixed weights instead of RL tuning	0.742	0.784	0.779	0.910	0.781	0.846
Without linear program (RL sets battery rates)*	0.763	0.862	0.849	–	0.854	–
Seasonal load and PV forecasts, 24 h	0.728	0.809	0.806	0.921	0.807	0.864
Untuned LightGBM, 24 h*	0.722	0.787	0.779	**0.908**	0.782	0.845
Horizon 24 h	0.723	**0.777**	0.772	0.913	**0.774**	**0.844**
Horizon 36 h	0.717	0.779	0.772	0.915	0.775	0.845
SAC instead of TQC (36 h)*	**0.715**	0.778	0.772	0.915	0.775	0.845

8th of 42, between 0.767 and 0.789 (Fig. 4). Our carbon score of 0.764 is the lowest of all 42 entries, 5% below the best one (0.803), for each of the three seeds, and our cost of 0.655 ranks 4th. Almost all of the difference to the winner lies in the grid KPI, where our D of 0.902 ranks 14th and the winner reaches 0.738.

The private leaderboard, which added the seven test buildings and decided the winners, was never published. The challenge overview reports only the participants' medians [13], $C = 0.792$, $G = 0.940$, $D = 0.996$ and a score of 0.907. With the same weights (0.2, 0.3, 0.5), our controller reaches $C = 0.728$, $G = 0.861$, $D = 0.942$ and 0.844, better than the median on every KPI. On the Phase I leaderboard (best entries 0.631, 0.635 and 0.651), our 0.717 would rank 5th.

E. Limitations of Load Forecasting

The load is by far the most difficult quantity to predict. With LightGBM, a single building's hourly load is 50–59% predictable (one minus the mean absolute error over the mean load), only slightly better than the seasonal rule. The district total is more predictable, at 71–78%, because the random variations of individual buildings partly cancel out, and this total is what the grid KPIs and the clipped cost see.

To find which errors cost score, we gave the fixed-weight controller the true future for one part of its forecasts at a time (Fig. 5), which bounds what better forecasts could give. With its own forecasts it scores $S = 0.784$ in the training and 0.779 in the validation district. Perfect PV forecasts barely change S (0.782 and 0.776), whereas perfect load forecasts lower it to 0.648 in both districts, well below the winner's level. Most of this value lies in the district total over the next few hours: rescaling the building forecasts to the true total gives 0.649 and 0.648, and the true load for only the next six hours gives 0.659 and 0.651. The performance of the controller is therefore limited mainly by the accuracy of the hourly district-load forecast, rather than by the solver or the tuner.

VII. DISCUSSION

In our design, RL is more useful as a tuner of the optimiser than as the controller itself. The forecasts and the linear program supply the daily patterns of price, PV and demand and the physics of the batteries, and the agent's three actions only adjust a schedule that already satisfies the battery constraints. The credit-assignment problem over thousands of hourly battery decisions thus becomes considerably simpler, which explains why removing the solver costs far more (public 0.854) than removing the tuner (0.781). Two remedies for the weaker grid KPI follow from the analysis: a load-factor term in the linear program, and better forecasts of the district total, which Section VI-E identifies as the main source of error.

Our study has several limitations. It covers one climate and one year of data, with five training buildings. The scores are computed with the official formulas on CityLearn 2.1.2 rather than with the challenge's evaluator, and most ablations use one seed. Beyond these results, the work serves several of the United Nations Sustainable Development Goals (SDGs), listed in Table III.

TABLE III
UNITED NATIONS SUSTAINABLE DEVELOPMENT GOALS ADDRESSED BY THIS WORK.

SDG	Justification
SDG 13 (Climate Action)	Reduces CO_2 emissions by 22%, the lowest carbon score of all leaderboard entries.
SDG 7 (Affordable and Clean Energy)	Reduces electricity costs by about a third and stores the rooftop PV surplus.
SDG 11 (Sustainable Cities and Communities)	Coordinates the batteries of a district and smooths the grid load.
SDG 9 (Industry, Innovation and Infrastructure)	Shows an AI control design for smart energy infrastructure.
SDG 12 (Responsible Consumption and Production)	Shifts consumption toward lower-price, lower-carbon hours.

VIII. CONCLUSION

This work proposed an RL-tuned model predictive controller in which LightGBM forecasts drive a receding-horizon linear program and a TQC agent retunes its weights every hour. On the CityLearn 2022 benchmark the controller reduced the electricity cost by 34.3% and the emissions by 22.2% in the training district and reached, over three seeds, a public score of 0.774 ± 0.003 and a private score of 0.844 ± 0.004, between 7th and 8th place on the public leaderboard, with the lowest carbon score of all 42 entries. Compared with the winner, the controller achieves a weaker grid KPI, and our analysis shows that this difference stems mainly from the accuracy of the district-load forecast; with perfect load forecasts the same solver would score about 0.65.

REFERENCES

[1] J. R. Vázquez-Canteli and Z. Nagy, "Reinforcement learning for demand response: A review of algorithms and modeling techniques," *Applied Energy*, vol. 235, pp. 1072–1089, 2019.

[2] Z. Wang and T. Hong, "Reinforcement learning for building controls: The opportunities and challenges," *Applied Energy*, vol. 269, p. 115036, 2020.

[3] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018.

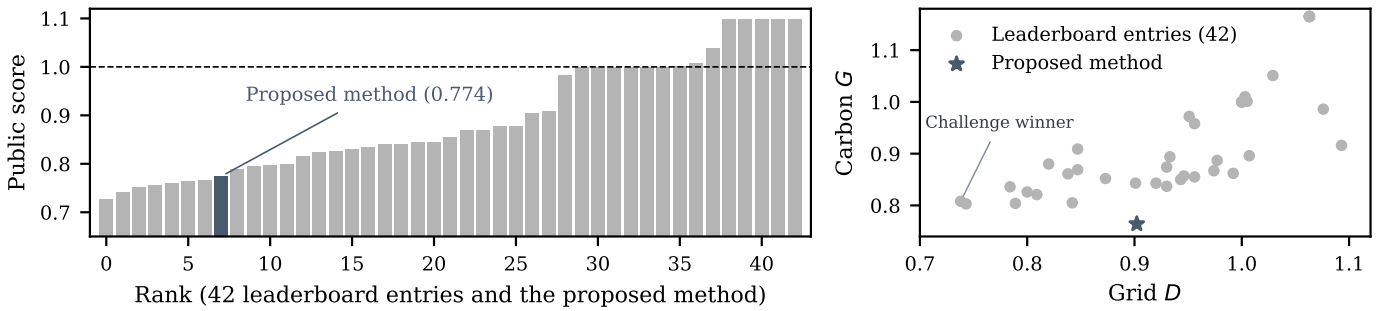


Fig. 4. The proposed method (dark bar and star) among the 42 entries of the Phase III public leaderboard. Left: public scores in rank order (dashed line: no battery). Right: carbon KPI against grid KPI. No entry achieves lower emissions, but most of the top-ranked entries achieve a lower grid KPI.

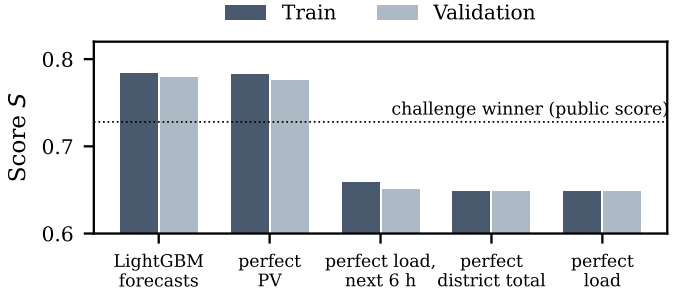


Fig. 5. Score of the fixed-weight controller when one part of its forecasts is replaced by the true future (an upper bound). The dotted line marks the winner's public score.

[4] L. Yu, S. Qin, M. Zhang, C. Shen, T. Jiang, and X. Guan, "A review of deep reinforcement learning for smart building energy management," *IEEE Internet of Things Journal*, vol. 8, no. 15, pp. 12 046–12 063, 2021.

[5] Z. Nagy *et al.*, "Ten questions concerning reinforcement learning for building energy management," *Building and Environment*, vol. 241, p. 110435, 2023.

[6] J. R. Vázquez-Canteli, J. Kämpf, G. Henze, and Z. Nagy, "CityLearn v1.0: An OpenAI Gym environment for demand response with deep reinforcement learning," in *Proc. 6th ACM Int. Conf. Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys)*, 2019, pp. 356–357.

[7] K. Nweye *et al.*, "CityLearn v2: Energy-flexible, resilient, occupant-centric, and carbon-aware management of grid-interactive communities," *Journal of Building Performance Simulation*, 2024.

[8] CityLearn Challenge Organizers, "The NeurIPS 2022 CityLearn challenge," AIcrowd, https://www.aicrowd.com/challenges/neurips-2022-citylearn-challenge, 2022.

[9] G. Dulac-Arnold, N. Levine, D. J. Mankowitz, J. Li, C. Paduraru, S. Gowal, and T. Hester, "Challenges of real-world reinforcement learning: Definitions, benchmarks and analysis," *Machine Learning*, vol. 110, no. 9, pp. 2419–2468, 2021.

[10] J. R. Vázquez-Canteli, G. Henze, and Z. Nagy, "MARLISA: Multi-agent reinforcement learning with iterative sequential action selection for load shaping of grid-interactive connected buildings," in *Proc. 7th ACM Int. Conf. Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys)*, 2020, pp. 170–179.

[11] K. Nweye, B. Liu, P. Stone, and Z. Nagy, "Real-world challenges for multi-agent reinforcement learning in grid-interactive buildings," *Energy and AI*, vol. 10, p. 100202, 2022.

[12] J. B. Rawlings, D. Q. Mayne, and M. M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed. Nob Hill Publishing, 2017.

[13] K. Nweye *et al.*, "The CityLearn challenge 2022: Overview, results, and lessons learned," in *Proc. NeurIPS 2022 Competitions Track*, ser. PMLR, vol. 220, 2023.

[14] T. P. Lillicrap, J. J. Hunt, A. Pritzel, N. Heess, T. Erez, Y. Tassa, D. Silver, and D. Wierstra, "Continuous control with deep reinforcement learning," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2016.

[15] S. Fujimoto, H. van Hoof, and D. Meger, "Addressing function approximation error in actor-critic methods," in *Proc. 35th Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 80, 2018, pp. 1587–1596.

[16] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *Proc. 35th Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 80, 2018, pp. 1861–1870.

[17] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

[18] A. Kuznetsov, P. Shvechikov, A. Grishin, and D. Vetrov, "Controlling overestimation bias with truncated mixture of continuous distributional quantile critics," in *Proc. 37th Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 119, 2020, pp. 5556–5566.

[19] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-Baselines3: Reliable reinforcement learning implementations," *Journal of Machine Learning Research*, vol. 22, no. 268, pp. 1–8, 2021.

[20] A. Y. Ng, D. Harada, and S. Russell, "Policy invariance under reward transformations: Theory and application to reward shaping," in *Proc. 16th Int. Conf. Machine Learning (ICML)*, 1999, pp. 278–287.

[21] J. Cao, D. Harrold, Z. Fan, T. Morstyn, D. Healey, and K. Li, "Deep reinforcement learning-based energy storage arbitrage with accurate lithium-ion battery degradation model," *IEEE Transactions on Smart Grid*, vol. 11, no. 5, pp. 4513–4521, 2020.

[22] P. Henderson, R. Islam, P. Bachman, J. Pineau, D. Precup, and D. Meger, "Deep reinforcement learning that matters," in *Proc. 32nd AAAI Conf. Artificial Intelligence*, 2018.

[23] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, and M. G. Bellemare, "Deep reinforcement learning at the edge of the statistical precipice," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021.

[24] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[25] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2019, pp. 2623–2631.